

# Package ‘LRg’

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**Title** Logistic Regression

**Version** 0.1.0

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**Description** Functions for fitting different logistic regression models. Models included are the multinomial logistic regression model, the parallel adjacent-categories logit model, and multi-dimensional stereotype ordered and unordered regression models.

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|             |                                        |
|-------------|----------------------------------------|
| auto_factor | <i>Automatic recoding into factors</i> |
|-------------|----------------------------------------|

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**Description**

Automatic recoding into factors

**Usage**

```
auto_factor(formula, data, max_levels = 10)
```

**Arguments**

|            |                                 |
|------------|---------------------------------|
| formula    | model expression                |
| data       | data frame                      |
| max_levels | maximum number of factor levels |

**Value**

recoded data frame

---

|      |                                 |
|------|---------------------------------|
| btls | <i>Backtracking line search</i> |
|------|---------------------------------|

---

**Description**

Backtracking line search

**Usage**

```
btls(x, X, Z, dx, u, alpha = 1, rho = 0.5, c = 1e-04, maxit = 20)
```

**Arguments**

|       |                              |
|-------|------------------------------|
| x     | parameter vector             |
| X     | design matrix                |
| Z     | response dummies             |
| dx    | search direction             |
| u     | score vector                 |
| alpha | step size                    |
| rho   | contraction factor           |
| c     | Armijo parameter             |
| maxit | maximum number of iterations |

**Value**

new parameter vector

---

|        |                                 |
|--------|---------------------------------|
| btlstn | <i>Backtracking line search</i> |
|--------|---------------------------------|

---

**Description**

Backtracking line search

**Usage**

```
btlstn(x, X, Z, q, dx, u, alpha = 1, rho = 0.5, c = 1e-04, maxit = 20)
```

**Arguments**

|       |                              |
|-------|------------------------------|
| x     | parameter vector             |
| X     | design matrix                |
| Z     | response dummies             |
| q     | rank                         |
| dx    | search direction             |
| u     | score vector                 |
| alpha | step size                    |
| rho   | contraction factor           |
| c     | Armijo parameter             |
| maxit | maximum number of iterations |

**Value**

new parameter vector

---

|           |                                   |
|-----------|-----------------------------------|
| cum_contr | <i>Cumulative binary recoding</i> |
|-----------|-----------------------------------|

---

**Description**

Cumulative binary recoding

**Usage**

```
cum_contr(x)
```

**Arguments**

|   |                |
|---|----------------|
| x | ordered factor |
|---|----------------|

**Value**

cumulative contrast

---

eyedisease

*Empirical dataset*

---

### Description

The data are 720 observations on 6 variables.

### Usage

eyedisease

### Format

A data frame with 720 rows and 6 columns

**diab** diabetes

**gh** glycosylated haemoglobin level

**dbp** diastolic blood pressure

**prot** proteinuria

**merl** maximum severity of retinopathy in the left and right eye

### Source

From the Wisconsin Epidemiological Study of Diabetic Retinopathy (Klein et al., 1984). From R package ordinalgmifs (Archer et al., 2014).

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jacob

*Jacobian*

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### Description

Jacobian

### Usage

jacob(alpha, gamma, Lambda, Phi)

### Arguments

alpha                vector of intercepts

gamma                vector of regressor effects

Lambda               matrix of regressor effects

Phi                   matrix of weights

### Value

Jacobian matrix

---

|           |                 |
|-----------|-----------------|
| jacob_stn | <i>Jacobian</i> |
|-----------|-----------------|

---

**Description**

Jacobian

**Usage**

jacob\_stn(Phi, Gamma)

**Arguments**

|       |        |
|-------|--------|
| Phi   | matrix |
| Gamma | matrix |

**Value**

Jacobian

---

|        |                                    |
|--------|------------------------------------|
| logLik | <i>The log-likelihood function</i> |
|--------|------------------------------------|

---

**Description**

The log-likelihood function

**Usage**

logLik(x, X, Z)

**Arguments**

|   |                  |
|---|------------------|
| x | parameter vector |
| X | desgin matrix    |
| Z | response dummies |

**Value**

log-likelihood value

---

|            |                                    |
|------------|------------------------------------|
| logLik_stn | <i>The log-likelihood function</i> |
|------------|------------------------------------|

---

**Description**

The log-likelihood function

**Usage**

```
logLik_stn(x, X, Z, q)
```

**Arguments**

|   |                  |
|---|------------------|
| x | parameter vector |
| X | design matrix    |
| Z | response dummies |
| q | rank             |

**Value**

log-likelihood value

---

|     |                              |
|-----|------------------------------|
| lrt | <i>Likelihood ratio test</i> |
|-----|------------------------------|

---

**Description**

Likelihood ratio test

**Usage**

```
lrt(obj1, obj2)
```

**Arguments**

|      |                     |
|------|---------------------|
| obj1 | fit results model 1 |
| obj2 | fit results model 2 |

**Value**

test statistic, df, and p-value

---

|     |                                        |
|-----|----------------------------------------|
| mlr | <i>Multinomial logistic regression</i> |
|-----|----------------------------------------|

---

**Description**

This function can be used to fit the multinomial logistic regression model.

**Usage**

```
mlr(formula, data, maxit = 100)
```

**Arguments**

|         |                              |
|---------|------------------------------|
| formula | model expression             |
| data    | data frame                   |
| maxit   | maximum number of iterations |

**Value**

A list of

|               |                                     |
|---------------|-------------------------------------|
| coefficients  | Parameter estimates                 |
| fitted.values | The model implied probabilities     |
| residuals     | Dummy responses minus fitted values |
| pred.cat      | The model implied categories        |

**Author(s)**

David J. Hessen

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|     |                                    |
|-----|------------------------------------|
| olr | <i>Ordinal logistic regression</i> |
|-----|------------------------------------|

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**Description**

This function can be used to fit the full rank ordinal logistic regression model using projected iterative weighted least squares.

**Usage**

```
olr(formula, data, maxit = 100)
```

**Arguments**

|         |                              |
|---------|------------------------------|
| formula | model expression             |
| data    | data frame                   |
| maxit   | maximum number of iterations |

**Value**

A list of

|                            |                                     |
|----------------------------|-------------------------------------|
| <code>coefficients</code>  | Parameter estimates                 |
| <code>fitted.values</code> | The model implied probabilities     |
| <code>residuals</code>     | Dummy responses minus fitted values |
| <code>pred.cat</code>      | The model implied categories        |

**Author(s)**

David J. Hessen

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|                  |                                                 |
|------------------|-------------------------------------------------|
| <code>pac</code> | <i>Parallel adjacent-categories logit model</i> |
|------------------|-------------------------------------------------|

---

**Description**

This function can be used to fit the parallel adjacent-categories logit model.

**Usage**

```
pac(formula, data, maxit = 100)
```

**Arguments**

|                      |                              |
|----------------------|------------------------------|
| <code>formula</code> | model expression             |
| <code>data</code>    | data frame                   |
| <code>maxit</code>   | maximum number of iterations |

**Value**

A list of

|                            |                                     |
|----------------------------|-------------------------------------|
| <code>coefficients</code>  | Parameter estimates                 |
| <code>fitted.values</code> | The model implied probabilities     |
| <code>residuals</code>     | Dummy responses minus fitted values |
| <code>pred.cat</code>      | The model implied categories        |

**Author(s)**

David J. Hessen



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`sim_data`*Example dataset*

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**Description**

A data set containing a single ordinal response variable and four continuous regressor variables. The response variable has four levels.

**Usage**`sim_data`**Format**

A data frame with 300 rows and 5 columns

**y** Response

**X1** Regressor variable 1

**X2** Regressor variable 2

**X3** Regressor variable 3

**X4** Regressor variable 4

**Source**

Simulated data

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`solve_safe`*Solving  $Idx=u$* 

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**Description**

Solving  $Idx=u$

**Usage**`solve_safe(I, u, lambda = 1e-08)`**Arguments**

`I` information matrix or Hessian

`u` score or gradient

`lambda` ridge penalty parameter

**Value**

ridge information matrix or Hessian

---

|     |                                      |
|-----|--------------------------------------|
| stn | <i>Stereotype nominal regression</i> |
|-----|--------------------------------------|

---

**Description**

This function can be used to fit a q-dimensional stereotype nominal regression model.

**Usage**

```
stn(formula, q = NULL, data, stv = NULL, csv = 0.001, tol = 1e-08, maxit = 100)
```

**Arguments**

|         |                              |
|---------|------------------------------|
| formula | model expression             |
| q       | rank                         |
| data    | data frame                   |
| stv     | vector of starting values    |
| csv     | common starting value        |
| tol     | convergence tolerance        |
| maxit   | maximum number of iterations |

**Value**

|                         |                                      |
|-------------------------|--------------------------------------|
| A list of               |                                      |
| alpha, Beta, Gamma, Phi | Parameter estimates                  |
| loglik                  | Value of the log-likelihood function |
| npars                   | Number of estimated parameters       |
| AIC                     | Akaike information criterion         |
| BIC                     | Bayesian information criterion       |
| fitted.values           | Model implied probabilities          |
| residuals               | Dummy responses minus fitted values  |
| pred.cat                | Model implied categories             |
| obs.cat                 | Observed categories                  |

**Author(s)**

David J. Hessen

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|     |                                      |
|-----|--------------------------------------|
| sto | <i>Stereotype ordered regression</i> |
|-----|--------------------------------------|

---

**Description**

This function can be used to fit a q-dimensional stereotype ordered regression model.

**Usage**

```
sto(formula, q = NULL, data, tol = 1e-08, maxit = 100)
```

**Arguments**

|         |                              |
|---------|------------------------------|
| formula | model expression             |
| q       | rank                         |
| data    | data frame                   |
| tol     | convergence tolerance        |
| maxit   | maximum number of iterations |

**Value**

|                         |                                      |
|-------------------------|--------------------------------------|
| A list of               |                                      |
| alpha, Beta, Gamma, Phi | Parameter estimates                  |
| loglik                  | Value of the log-likelihood function |
| npars                   | Number of estimated parameters       |
| AIC                     | Akaike information criterion         |
| BIC                     | Bayesian information criterion       |

**Author(s)**

David J. Hessen

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